

Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.



Project Overview

1

Analyze Behavior

3,900 purchases across categories.

2

Uncover Insights

Spending patterns, segments, preferences.

3

Guide Decisions

Optimize strategy with data.

Dataset Summary

Data Points

- Rows: 3,900
- Columns: 18

Key Features

- Customer demographics
- Purchase details
- Shopping behavior



Missing Data

37 values in Review Rating column.

Exploratory Data Analysis (Python)

Data Loading & Exploration

Imported with pandas, checked structure.

Missing Data Handling

Imputed Review Rating using median.

Column Standardization

Renamed to snake_case.

Feature Engineering

Created age_group, purchase_frequency_days.

Database Integration

Loaded into PostgreSQL for SQL analysis.

Data Analysis (SQL)

Key business questions answered through PostgreSQL.

1

Revenue by Gender

Female: \$75,191, Male: \$157,890

2

High-Spending Discount Users

839 customers identified.

3

Top 5 Products by Rating

Gloves, Sandals, Boots, Hat, Skirt.

```
1  vual crata alent   
2  /tal enolent/can));{   
3  ()   
6  ripe (batrmoleb;   
7  / (Lan);   
8  wat (tp);   
4  certataddl((RateIarIng(assich();   
15  watf );   
15  mualcital axpe.(intaless/)   
  
12  fix//umush(Butle ras/actead;   
13  );   
13  ororage letvel(Apblan east);   
15  mdatabaseI(halckin == lecaling(fona/(dat ablests));   
  
16  fls//uane(dtalbases(reIenstuotes);   
15  );   
15  detfante(coatcases(lips);   
15  
```

SQL Analysis: Shipping & Subscriptions

Shipping Type Comparison

- Standard: \$58.46
- Express: \$60.48

Subscribers vs. Non-Subscribers

- Subscribers: 1,053, Avg Spend: \$59.49
- Non-Subscribers: 2,847, Avg Spend: \$59.87



SQL Analysis: Products & Segments

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%).

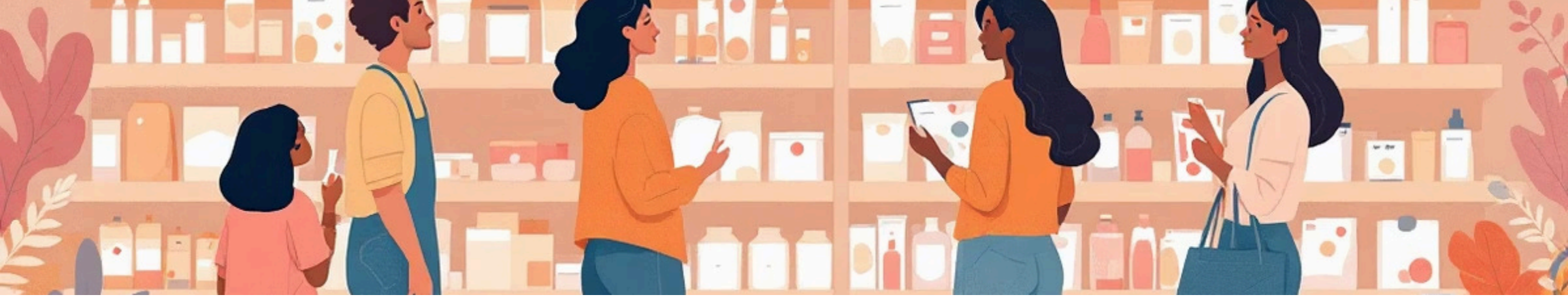
Customer Segmentation

Loyal: 3,116, Returning: 701, New: 83.

Repeat Buyers & Subscriptions

958 repeat buyers are subscribers.





Top Products & Revenue by Age

Top 3 Products per Category

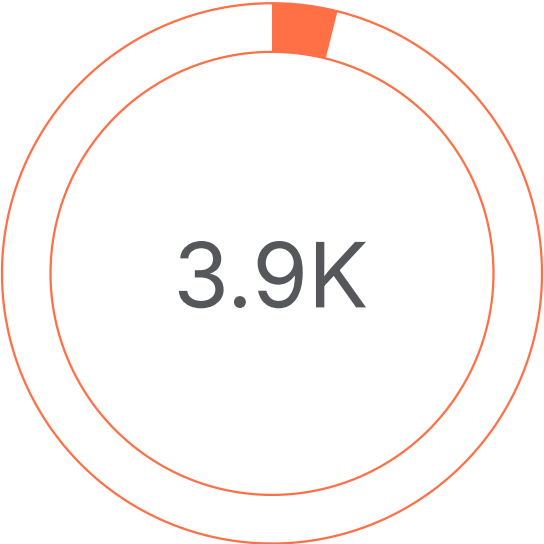
- Accessories: Jewelry, Sunglasses, Belt
- Clothing: Blouse, Pants, Shirt
- Footwear: Sandals, Shoes, Sneakers
- Outerwear: Jacket, Coat

Revenue by Age Group

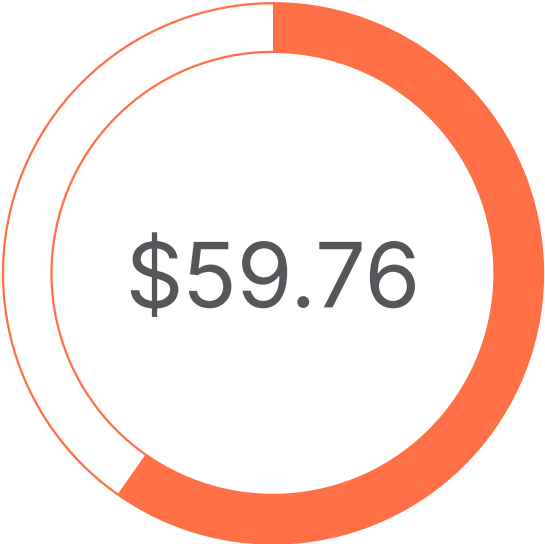
- Young Adult: \$62,143
- Middle-aged: \$59,197
- Adult: \$55,978
- Senior: \$55,763

Power BI Dashboard

Interactive visualizations for key insights.



Customers



Avg Purchase



Avg Review

Business Recommendations

→ Boost Subscriptions

Promote exclusive benefits.

→ Customer Loyalty

Reward repeat buyers.

→ Review Discount Policy

Balance sales with margins.

→ Product Positioning

Highlight top-rated items.

→ Targeted Marketing

Focus on high-revenue groups.